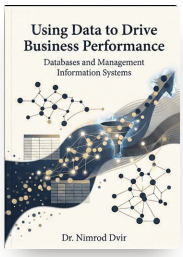
Using Data to Drive Business Performance

Databases and Management Information Systems

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Version 1 tested in summer 2026 · **Version 2** in use with 500 students in fall 2026

What is this book about?

Courseware that teaches business students to turn data into decisions: a textbook, interactive labs, auto-graded quizzes, guided practice and a semester project, built to run as one course. Students learn to organize data, query it with SQL, check the answers, present them and decide, with AI as an assistant they learn to check.

Data → Tables → Relationships → Queries → Analytics → Decisions

Who it is for	The required business-core data or IS course in any major, the MIS database course, and adjacent courses such as accounting information systems.

Why and how it was built

I wrote this book because I could not find one that matched my needs as cloud databases, analytics and AI changed the field. I saw a strong market opportunity and wanted full control over the content.

Seven years in the classroom	Developed while teaching the introductory MIS course at the University at Albany to more than 4,000 students.
A decade of editing	Ten years as a magazine editor shaped clear, visual writing at a level business undergraduates can follow.
Grounded in research	63 syllabi from 31 institutions, labor-market data and a chapter-level comparison of the leading textbooks.

Already in version 2. Tested in summer 2026 with a small group of students, and again in fall 2026 with 500 students. The full digital edition is live with an instructor dashboard.

Market research

WHAT IT MEASURES	FIGURE	SOURCE
Business bachelor's degrees awarded in the U.S. each year (2021–22)	**375,418**	NCES Digest, table 318.20
Business schools whose core requires an information systems course (61 schools)	**88%**	Fustos et al., 2017
Estimate: business students reaching a required IS course each year	**About 330,000**	375,418 × 88%
Estimate: MIS students in programs with a database course each year	**About 10,500**	12,092 MIS degrees (Data USA) × 87.3% of IS programs (Hwang et al., 2015)
U.S. data analyst job postings that list SQL (Power BI 29%)	**52.9%**	365 Data Science, 2026
Projected growth in data scientist jobs, 2025–35 (all occupations +3%)	**+35%**	U.S. BLS
Textbooks named in the 63 syllabi; none used at more than 3 of 31 schools	**60 titles**	Our syllabus research

How it is different

This book teaches business students to use data to make decisions. Therefore, it differs from the leading textbooks mainly by:

- **Starting with the business.** Chapter 1 opens with performance, goals and KPIs, before any database theory.
- **Teaching data before databases.** Data types, measurement scales and organizing information come first.
- **Using a database before designing one.** Students query a supplied database, then design their own.
- **Ending in a decision.** Analytics, dashboards and a full Power BI chapter lead to a decision memo.
- **One SQL, flexible tools.** One PostgreSQL dialect, with tested MySQL and Access routes on the same cases.
- **Making AI a habit.** Every chapter has an AI Check: accept, revise or reject AI-written work.

Sources: NCES; Fustos et al. (2017); Hwang, Ma & Wang (2015), ISEDJ 13(6); Data USA; 365 Data Science (2026); U.S. BLS; author syllabus research. Full references in the online prospectus.

Suggested outline

A preface, 16 chapters in five parts, a midterm review week, a closing chapter on careers and seven appendices. It fits a 14-week semester. Business-core route: Chapters 1 to 8 and 12 to 14. MIS database route: Chapters 1 and 4 to 16.

PART I From Data to Decisions

1 From Data to Business Performance
Business, KPIs, DIKW, information behavior, IS and IT

2 Understanding Data
Data types, measurement scales, CSV and JSON, data quality

3 Organizing Data and Information
Files, metadata, tags, taxonomies, data dictionaries

PART II Databases and SQL

4 Databases and the Relational Model
DBMS, keys, relationships, schemas, integrity

5 SQL Foundations
SELECT to aggregates, CREATE and INSERT, first AI Check

6 SQL for Business Analysis
Joins, GROUP BY, CASE, business measures

MID Integration Studio A
Review, practice exam, project checkpoint 1

PART V Putting It to Work

16 Capstone: From Data to Decisions
Problem to database to dashboard to decision memo

END Closing: Where Data Can Take You
Data careers and building a portfolio

PART III Designing and Running Databases

7 Database Design: Requirements to ER Diagrams
Business rules, Crow's Foot notation, Lucidchart

8 Normalization
Anomalies, dependencies, 1NF to 3NF

9 Building and Testing a Database
DDL, constraints, loading data, integrity tests

10 Advanced SQL
CTEs, views, window functions, triggers

11 Administration, Security and Reliability
Roles, row-level security, transactions, backup

PART IV From Data to Insight

12 Business Analytics and Decision-Making
Fair comparisons, causation, the decision memo

13 Reporting and Data Visualization
Dashboards, chart choice, honest visuals

14 Power BI
Power Query, data model, DAX measures

15 Big Data, Data Warehouses and AI
Star schemas, ETL, NoSQL, verifying AI output

APPENDICES Tools and routes

A Supabase · B MySQL route · C Access route · D SQL dialect guide · E Lucidchart · F Tableau route · G Terms Treasury

Every chapter: Introduction, Core Concepts, Terms Treasury, Review Questions, an auto-graded Reading Assessment Test, Let's Build, an Interactive Lab and Key Takeaways.

Advantages of this courseware

Ready asset	Live, in version 2 and used by 500 students this semester. Textbook, labs, quizzes and instructor materials exist today.
Two markets	One book serves the business-core data course and the MIS database course through two routes.
Open slot	No textbook dominates this course. Cengage's applied title, *Concepts of Database Management*, was last revised in 2021.
Known partner	I already work with Cengage on its Intro Computing Thought Leadership Board.
Additional opportunities	Adjacent courses, custom editions, and MySQL and Access routes that let current adopters keep their labs.

There are a few questions I would like to discuss.

1 **Positioning and content:** review the outline together: what to add, where to complement the current books, where not to compete.

2 **Tools and access:** which routes to adopt and which SQL platforms to support (MS Access, MySQL, PostgreSQL); Power BI for reporting.

3 **Platform fit:** how the chapter package and a live SQL environment map to MindTap.

4 **Running cases:** keep the grading database and PetVax, or add a commercial case?

5 **Review package:** what should be included in the review, how many reviewers, and with what criteria.

Suggested next steps

1 Decide whether we move forward, and cement the relationship.

2 Schedule a call to review the open questions and the suggested outline together, and decide what to focus on.

3 Develop the manuscript and submit it.

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